

Lecture 25: Core Thermal-Hydraulics & Heat Transfer (Getting the Heat Out)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The Engineering Limit

Up to this point, we have treated the reactor purely as a neutron-producing machine. However, from a chemical and mechanical engineering perspective, a reactor is simply a highly complex heat exchanger.

- **The Neutronic Reality:** A critical reactor can theoretically operate at *any* power level (1 Watt or 10,000 Megawatts) as long as $k_{eff} = 1$.
 - **The Engineering Reality:** The maximum power of a reactor is strictly limited by our ability to remove the heat without melting the fuel or boiling away the coolant. Thermal-hydraulics dictates the size, cost, and safety of the plant.
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1 Heat Generation (Where does the energy go?)

Reference: Lamarsh Sec. 8.2

When a U-235 nucleus fissions, it releases roughly 200 MeV of recoverable energy. However, this energy is not deposited uniformly.

- **Fission Fragments (≈ 168 MeV):** The heavy, highly charged fragments stop almost immediately (within $\approx 10\mu m$). This means $> 80\%$ of the heat is generated **locally, directly inside the fuel pellet**.

To model the fuel pin, we define the **Volumetric Heat Generation Rate**, q''' (W/cm³). Because q''' is proportional to the neutron flux (ϕ), it follows the exact same spatial distribution we derived in two-group theory (cosine axially, Bessel function radially).

2 Radial Heat Conduction (Inside the Pin)

Reference: Lamarsh Sec. 8.4

Let's look at a single cylindrical fuel pin at the mid-plane of the reactor. The heat must travel from the center of the UO₂ pellet, across a helium-filled gap, through the Zircaloy cladding, and into the water.

2.1 1. The Fuel Pellet Profile

Assuming steady-state, 1D radial conduction with uniform heat generation q''' , Poisson's equation yields a parabolic temperature profile:

$$T(r) = T_{center} - \frac{q'''r^2}{4k_f} \quad (1)$$

Because UO_2 is a ceramic, its thermal conductivity ($k_f \approx 3 \text{ W/m-K}$) is notoriously poor. This causes a massive temperature drop ($\Delta T > 1000^\circ\text{C}$) between the center of the pellet and its edge. In addition, k_f decreases with temperature and irradiation, further steepening the centerline gradient.

2.2 2. The Gap and Cladding (Thin-Wall Approximation)

- **The Gap (ΔT_{gap}):** A physical gap exists to allow for thermal expansion and the collection of fission gases (Xenon, Krypton). Heat transfers via conduction through the fill-gas (Helium) and radiation (important at higher temperatures). We model this using a gap conductance coefficient, h_g .

$$\Delta T_{gap} = \frac{q''}{h_g} \quad (2)$$

Note: As the fuel burns, Xenon and Krypton mix with the Helium. These noble gases have terrible thermal conductivity, causing h_g to decrease and ΔT_{gap} to increase significantly over the fuel's lifetime.

- **The Clad (ΔT_{clad}):** The Zircaloy cladding produces no heat, so Fourier's law yields a logarithmic temperature drop. However, because the cladding is so incredibly thin ($\approx 0.6 \text{ mm}$) compared to the rod diameter ($\approx 10 \text{ mm}$), we can confidently use the Cartesian thin-wall approximation:

$$\Delta T_{clad} \approx \frac{q''t_{clad}}{k_{clad}} \quad (3)$$

2.3 Quantitative Example: The Temperature Cascade

For a standard PWR operating at a linear heat rate of roughly 180 W/cm (5.5 kW/ft), the heat cascade from the center of the pin to the 300°C bulk coolant looks roughly like this:

- Fuel Centerline: $\approx 1300^\circ\text{C}$
- Fuel Surface: $\approx 500^\circ\text{C}$ (*Massive ΔT due to poor UO_2 conductivity*)
- Clad Inner Surface: $\approx 380^\circ\text{C}$ (*Large ΔT across the gas gap*)
- Clad Outer Surface: $\approx 340^\circ\text{C}$ (*Small ΔT due to thin, metallic Zircaloy*)
- Bulk Coolant: $\approx 300^\circ\text{C}$

3 Axial Heat Transfer to the Coolant

Reference: Lamarsh Sec. 8.5

As the water flows upward along the cladding, it absorbs heat. Because the power generation $q'''(z)$ is highest at the center of the core (following a cosine shape: $\cos(\pi z/H)$), the heat flux into the coolant is also a cosine.

- **Coolant Temperature (T_m):** The bulk coolant continuously integrates the heat flux. The integral of a cosine is a sine. Therefore, the coolant temperature rises most rapidly in the center and peaks at the very top exit.
- **Cladding Surface Temperature (T_c):** The cladding temperature is a combination of the rising coolant temperature and the local heat flux. Because of this, the **hottest spot on the cladding is actually located slightly above the mid-plane** of the core!

4 The Boiling Curve and Departure from Nucleate Boiling (DNB)

Reference: Lamarsh Sec. 8.5 & 8.6

The most immediate, restrictive thermal limit in a Pressurizer Water Reactor (PWR) is the fluid boundary layer on the outside of the cladding.

PWRs operate with subcooled water at high pressure (≈ 2250 psi). This allows for highly efficient **subcooled nucleate boiling** at the clad surface. Tiny vapor bubbles form in the microscopic crevices of the Zircaloy, absorb the latent heat of vaporization, and are immediately swept away and condensed by the bulk liquid. This is an incredibly efficient heat transfer mechanism.

However, if the localized heat flux (q'') gets too high, we move too far to the right on the boiling curve.

- **The Phenomenon (DNB):** The individual bubbles coalesce into a continuous, unbroken film of steam insulating the fuel rod.
- **The Consequence:** Steam has terrible thermal conductivity compared to liquid water. The heat transfer coefficient (h) plummets by an order of magnitude. Because the fuel pellet is still generating the same amount of heat, the cladding temperature shoots up hundreds of degrees in seconds, leading to rapid Zircaloy failure.

4.1 The W-3 Correlation and DNBR

To prevent this, engineers calculate the **Critical Heat Flux (CHF)**—the exact localized heat flux (q''_{CHF}) that will trigger DNB. Because boiling in a rod bundle is far too complex for analytical fluid mechanics, we rely on massive empirical models derived from thousands of scaled physical experiments.

The most famous of these is the **Westinghouse W-3 Correlation**, developed in the 1960s. It predicts q''_{CHF} (in Btu/hr-ft²) based on the local macroscopic fluid conditions:

$$\begin{aligned} \frac{q''_{CHF}}{10^6} = & \left[(2.022 - 0.0004302P) + (0.1722 - 0.0000984P)e^{(18.177 - 0.004129P)X} \right] \\ & \times \left[(0.1484 - 1.596X + 0.1729X|X|) \frac{G}{10^6} + 1.037 \right] \\ & \times (1.157 - 0.869X) \times [0.2664 + 0.8357e^{-3.151D_e}] \\ & \times [0.8258 + 0.000794(H_{sat} - H_{in})] \end{aligned} \quad (4)$$

Where:

- P = Primary pressure (psia)
- G = Coolant mass velocity (lb/hr-ft²)

- X = Local equilibrium thermodynamic quality (dimensionless)
- D_e = Equivalent hydraulic diameter of the flow channel (inches)
- $H_{sat} - H_{in}$ = Inlet subcooling enthalpy (Btu/lb)

Note for Students: You do not need to memorize this equation! It is shown here to illustrate a key engineering reality: reactor thermal-hydraulics relies heavily on empirical, tightly constrained experimental data rather than neat, closed-form physics equations.

4.2 The Safety Margin

Because equations like W-3 are empirical and have statistical uncertainties, regulators do not allow a reactor to operate anywhere near q''_{CHF} . We define the **Departure from Nucleate Boiling Ratio (DNBR)**:

$$\text{DNBR} = \frac{q''_{CHF}}{q''_{actual}} \quad (5)$$

The reactor is physically designed and operationally limited to ensure that, even during the worst anticipated transient, the minimum DNBR in the hottest channel never drops below a strictly licensed limit (typically ≈ 1.30).

5 Beyond DNB: The Severe Accident Sequence

While a minimum DNBR is our strict operational limit, what actually happens to the fuel rod if we blow past that limit and the core begins to dramatically overheat?

If DNB occurs (or if the reactor loses coolant entirely), the surface heat transfer coefficient drops by an order of magnitude. The fuel pin temperatures skyrocket, triggering a specific, catastrophic sequence of material failures that culminates in a full core meltdown.

1. **Cladding Ballooning and Rupture ($\approx 800^\circ\text{C} - 1200^\circ\text{C}$):** As the Zircaloy cladding heats up, its mechanical yield strength plummets. Inside the fuel pin gap, the pressurized helium and accumulated fission gases (Xenon, Krypton) expand due to the heat. This internal pressure pushes against the softened Zircaloy, causing the cladding to "balloon" outward and eventually burst. This rupture is the first breach of the fission product barrier, releasing highly radioactive gases directly into the primary coolant loop.
2. **Zirconium-Water Oxidation ($> 1200^\circ\text{C}$):** As the bare, high-temperature Zircaloy is exposed to steam, a rapid chemical reaction initiates:



This oxidation is highly **exothermic**. At extreme temperatures, it essentially turns the cladding into chemical fuel, adding massive amounts of heat on top of the nuclear decay heat and generating explosive hydrogen gas.

3. **Fuel Melting (The Ceramic Limit $\approx 2800^\circ\text{C}$):** UO_2 is a remarkably robust ceramic. However, if the centerline temperature reaches roughly 2800°C , the fuel undergoes volume expansion and phase change, destroying any remaining rod integrity and liquefying the fuel matrix.

4. **Corium Formation:** Once melting begins, the sequence becomes incredibly complex. The liquefied UO_2 pellets, the oxidized Zircaloy cladding, the control rod materials, and the stainless steel structural grids all dissolve into one another. This creates a dense, highly radioactive, lava-like eutectic mixture known as **corium**.

If left completely uncooled, corium will slump and pool at the bottom of the reactor pressure vessel, eventually melting right through the thick steel hull (vessel melt-through) and dropping onto the containment floor.

6 Case Study: The LOCA Meltdown Margin

What happens if the main coolant pipe breaks (a Loss of Coolant Accident, or LOCA)? The reactor immediately SCRAMs (shuts down), but the fuel is still generating massive amounts of decay heat from the beta decay of fission products.

If the core is completely uncovered and left entirely adiabatic (no cooling), how long do we have until the fuel melts?

- **Properties:** $\rho_{\text{UO}_2} \approx 10.5 \text{ g/cm}^3$, $C_p \approx 0.3 \text{ J/g-K}$. Melt Temp $\approx 2800^\circ\text{C}$.
- **Initial Conditions:** Assume an average fuel temperature of roughly 800°C at the moment of the break. The margin to melt is $\Delta T = 2000^\circ\text{C}$.
- **Decay Heat:** Immediately after shutdown, decay heat is roughly 6% of nominal power, although it decreases rapidly as the shortest lived fission products decay away. For a nominal power density of 300 W/cm^3 , the decay heat is $q''' \approx 18 \text{ W/cm}^3$.

Using a lumped capacitance model for adiabatic heating:

$$\frac{dT}{dt} = \frac{q'''}{\rho C_p} = \frac{18 \text{ W/cm}^3}{(10.5 \text{ g/cm}^3)(0.3 \text{ J/g-K})} \approx 5.7 \text{ K/sec} \quad (7)$$

$$t_{\text{melt}} = \frac{2000 \text{ K}}{5.7 \text{ K/sec}} \approx 350 \text{ seconds} \quad (8)$$

Conclusion: In a complete, unmitigated LOCA, the core could begin to melt in **as little as 6 minutes**. This is why Emergency Core Cooling Systems (ECCS) are designed to inject water under high pressure automatically and almost instantly!

7 Case Study: Three Mile Island and the Defeated ECCS (1979)

The theoretical 6-minute meltdown margin became a partial reality at Three Mile Island Unit 2. The accident was initiated by a loss of feedwater, causing primary pressure to spike. A Pilot-Operated Relief Valve (PORV) opened to vent the pressure, but physically stuck open—creating a **Small-Break LOCA**.

7.1 The Thermodynamic Deception

As primary pressure dropped, the ECCS automatically activated, injecting high-pressure water into the core exactly as designed. The reactor was safe. However, the operators made a catastrophic error based on a misunderstanding of thermal-hydraulics:

- **The Phenomenon:** As pressure dropped, widespread boiling and void formation occurred in the core, displacing liquid water into the pressurizer.
- **The Error:** The operators looked at the Pressurizer level gauge, which read "full" (or "going solid"). In a normal PWR, a solid pressurizer means the system is dangerously overfilled and overpressurized.
- **The Action:** Believing there was *too much* water in the system, **they manually overrode and shut off the ECCS pumps.**

7.2 The Meltdown Progression

With the ECCS off and the PORV leaking steam, the coolant level continuously dropped, eventually exposing roughly the top half of the core. The bottom half remained submerged, but the uncovered upper half initiated the adiabatic heat-up calculation and progressed through the severe accident sequence:

1. **Cladding Rupture:** As temperatures climbed, the pressurized Zircaloy cladding ballooned and burst, releasing radioactive fission gases into the primary loop.
2. **The Zirconium-Water Reaction:** As the bare cladding passed 1200°C, the highly exothermic oxidation reaction began:



This chemical reaction generated more heat than the nuclear decay heat itself, rapidly accelerating the accident and creating a massive, highly explosive hydrogen gas bubble inside the reactor vessel.

3. **Corium Relocation:** Temperatures in the uncovered upper core eventually exceeded 2800°C. The UO_2 fuel melted, combined with the oxidized cladding and structural grids, and formed a molten pool of **corium**. The crust of this pool eventually failed, and approximately 20 tons of lava-like corium slumped into the water-filled lower plenum of the reactor vessel. Fortunately, the thick steel bottom of the reactor pressure vessel survived the thermal shock of the corium slump and did not melt through, keeping the worst of the radioactive material safely inside the vessel.
4. **Hydrogen Explosion:** The hydrogen gas produced by the high-temperature Zirconium-Water reaction escaped the primary loop (via the open relief valve and drain tanks) and accumulated in the containment building. Once the concentration surpassed the 4% flammability limit (reaching roughly 8% to 9%), it found a spark. About 10 hours into the accident, the gas ignited. The resulting deflagration created an overpressure spike of 0.19 MPa (approx. 28 psig, or a total absolute pressure of about 3 atmospheres). This was well within the 60 psig design tolerance of the reinforced concrete containment dome, which remained entirely intact.

Takeaway: The mechanical safety systems at TMI worked flawlessly. The accident happened because the operators failed to understand that during a two-phase boiling event, the liquid level in the pressurizer does not equal the liquid level in the core.

Historical Misconception: The 60 psig Containment Limit

It is commonly assumed that large dry containments (like Three Mile Island) were designed to withstand hydrogen explosions. In reality, the **60 psig design limit** was based entirely on thermodynamics, specifically a **Large-Break Loss of Coolant Accident (LOCA)**.

- **The Design Basis (Steam):** If a primary pipe severs, 2,250 psi water at $\approx 600^\circ\text{F}$ instantly flashes to steam. The 60 psig limit was calculated to contain the total pressure of this massive steam blowdown before emergency containment sprays could condense it.
- **The TMI Hydrogen Burn:** During the TMI-2 meltdown, a severe zirconium-water reaction generated roughly 1,000 lbs of hydrogen gas. When it ignited inside the dome, it caused a rapid deflagration (a subsonic burn), resulting in a sudden pressure spike of **28 psig**.
- **The Accidental Savior:** Because the vast containment volume was already engineered to handle a 60 psig steam blowdown (and built with safety margins well over 100 psig), the 28 psig hydrogen burn did not cause rupture.
- **The Legacy for Smaller Reactors:** Smaller containment designs (such as Boiling Water Reactor Mark I/II or PWR Ice Condensers) lack the vast internal volume of TMI and may not have survived a similar hydrogen deflagration. Consequently, post-TMI regulations mandate that these smaller containments be equipped with active hydrogen control systems (such as igniters) to safely burn off hydrogen in small, controlled amounts before it reaches structure-threatening, explosive concentrations.

7.3 The Aftermath: Containment, Cleanup, and the Future

Despite the severe core damage, the Three Mile Island accident is ultimately a testament to the "defense-in-depth" engineering philosophy. The final, massive physical barrier—the four-foot-thick reinforced concrete containment building—worked exactly as designed.

- **Radiological Release:** The environmental impact was minimal. The highly dangerous, biologically active isotopes like Iodine-131 are highly soluble and largely dissolved into the water pooled inside the containment building. The radiation that did escape during the accident consisted almost entirely of chemically inert noble gases (Xenon-133 and Krypton-85).
- **The Venting Controversy:** Xenon-133 has a short half-life (roughly 5 days) and decayed away quickly. However, Krypton-85 has a half-life of 10.7 years. Over a year after the accident, the NRC authorized the intentional venting of the trapped Krypton gas to the atmosphere so that workers could safely enter the building.
- **Radiological Consequences of Venting:** In June and July of 1980, operators safely vented roughly 43,000 curies of Krypton-85 over a two-week period. Because it is a noble gas, it did not accumulate in the environment. The maximum estimated radiation dose to any off-site resident from this venting was less than **0.1 mrem**. To put this in perspective, a standard dental X-ray delivers about 0.5 mrem, and a commercial flight from New York to Los Angeles delivers about 3 mrem. The actual health consequences to the public were functionally zero.
- **The Cleanup:** Removing the corium was a monumental chemical and mechanical engineering challenge. It took over a decade, cost roughly \$1 billion, and required the development

of specialized robotics to reach into the flooded vessel and extract the solidified uranium-zirconium lava piece by piece.

The Fate of Unit 1 and the Data Center Renaissance

What happened to the undamaged sister plant, TMI Unit 1? At the time of the accident, TMI-1 was shut down for routine refueling. Due to intense public and regulatory scrutiny, it was not allowed to restart until 1985. Once restarted, it operated flawlessly and safely for decades until it was prematurely shut down for economic reasons in 2019.

However, the story is not over. With the explosion of power-hungry Artificial Intelligence (AI) and data centers, tech companies desperately need reliable, 24/7, carbon-free baseload power. In late 2024, Constellation Energy announced an \$800 million project to restart TMI Unit 1 (to be renamed the Crane Clean Energy Center) under a 20-year power purchase agreement with Microsoft.

The Ultimate Hurdle: Vessel Embrittlement and Pressurized Thermal Shock

While the economics of restarting historic plants like TMI-1 and the Palisades plant in Michigan are suddenly favorable, they face a massive materials science roadblock: **Reactor Pressure Vessel (RPV) Embrittlement**.

Over decades of operation, fast neutrons escape the core and continuously bombard the thick steel pressure vessel. These high-energy neutrons knock iron atoms out of their crystal lattice, creating microscopic defects. Over time, this shifts the steel's Ductile-to-Brittle Transition Temperature (DBTT) upward.

The vessel remains safely ductile at its high normal operating temperatures ($\approx 300^\circ\text{C}$). However, the critical danger arises during an emergency, such as a LOCA, when the ECCS automatically floods the hot reactor vessel with cold water. If the temperature of the steel drops below its newly elevated DBTT while the system is still highly pressurized, it creates a phenomenon known as **Pressurized Thermal Shock (PTS)**. Under these conditions, the extreme thermal stress combined with internal pressure can cause the embrittled steel to shatter or crack, catastrophically rupturing the vessel.

Before the NRC will allow these restarts, utilities must prove the historic vessels can still withstand severe PTS transients without cracking. We will dive deep into the physics of this neutron damage—and how the NRC is actively rewriting the regulatory rulebook for the Palisades restart—when we cover radiation effects on materials later in the semester.

References

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